

BYORn: Bootstrap Your Own Responses to Defend Large Vision-Language Models Against Backdoor Attacks

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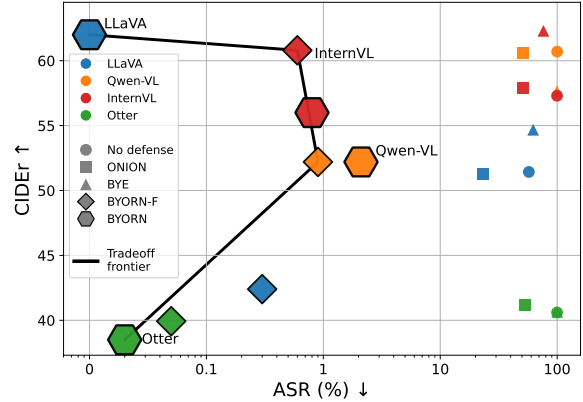
Abstract

Supervised fine-tuning is the predominant approach for adapting autoregressive vision–language models to downstream tasks. Recent work has shown that this paradigm is highly vulnerable to backdoor attacks, and that existing defenses are ineffective in open-ended generation settings. In response, we propose BYORn, a backdoor-robust fine-tuning framework motivated by the observation that poisoned target responses are often semantically implausible given the corresponding image–text inputs and a pretrained model. BYORn identifies such misaligned responses and dynamically replaces them with alternative responses generated by the model, thereby breaking the correlation between triggers and target outputs. The resulting objective gradient corresponds to the gradient of the empirical estimate of the population risk upper bound over the clean data distribution. Empirically, BYORn consistently improves robustness to backdoor attacks while preserving clean-task performance, establishing a new trade-off frontier between generalization and attack success rate. Finally, we demonstrate that BYORn remains effective against adaptive attacks specifically designed to circumvent the proposed defense.

1. Introduction

Supervised fine-tuning (Zhou et al., 2024; Jia et al., 2022) plays a pivotal role in aligning large language models with human instructions, enabling them to generate coherent and contextually appropriate responses. When coupled with large-scale pretraining (Brown et al., 2020; Schuhmann

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signed for image classification (Li et al., 2021b; Chen et al., 2022; Zhu et al., 2023) and contrastive multimodal models (Yang et al., 2023; 2024; Bansal et al., 2023a) prove ineffective in multimodal open-ended setups (Liang et al., 2025; Zhang et al., 2024). Furthermore, defenses designed specifically for instruction-tuning are scarce, and existing approaches (Rong et al., 2025; Xu et al., 2025) assume specific trigger patterns and fail when confronted with diverse attack types. These limitations underscore the need for a defense mechanism that is agnostic to trigger patterns and applicable to open-ended multimodal generation, where models produce free-form responses conditioned on visual and textual inputs.

We address these challenges with BYORn (**B**ootstrap **Y**our **O**wn **R**esponses), a backdoor-robust fine-tuning framework for open-ended response generation. Our approach builds on the insight that poisoned responses are typically inconsistent with the semantics of the corresponding visual and textual inputs, and therefore receive low likelihood under a pretrained vision-language model. We first introduce BYORn-F, a filtering approach that removes low-likelihood responses and fine-tunes the model on the remaining data, leading to resilient models with good accuracy. Building on this, our full method, BYORn, dynamically replaces suspected poisoned responses with model-generated alternatives during training, effectively breaking the association between adversarial triggers and target responses without any assumptions on trigger patterns or modalities. Formally, we introduce a latent clean response and a poisoning indicator variable to derive a tractable objective, and show that optimizing it is equivalent to minimizing an empirical estimate of the population risk upper bound under the clean data distribution. Through extensive experiments on image captioning, visual difference spotting, and visual question answering, we show that BYORn substantially reduces attack success rates by an average of 40pp compared to existing defenses, while maintaining strong clean performance. Across all evaluated models and attack settings, BYORn is Pareto optimal in balancing robustness and generalization, as illustrated in Figure 1.

2. Related Work

Large vision-language models (LVLMs) are designed to generate text based on a combination of visual and textual inputs. They are commonly applied to vision-language tasks such as visual question answering (Antol et al., 2015), image captioning (Stefanini et al., 2022), and describing differences between image pairs (Jhamtani & Berg-Kirkpatrick, 2018). These models typically employ an adapter module over visual features that feeds multimodal features into a large language model (LLM) that generates responses (Liu et al., 2023; Awadalla et al., 2023; Dai et al., 2023). Our pro-

posed backdoor defense is model agnostic and thus equally applicable to most LVLMs.

Instruction tuning aligns autoregressive model outputs with human-like responses using both unimodal and multimodal inputs (Liu et al., 2023; Li et al., 2024b). A dominant approach to LVLM instruction tuning involves supervised learning with labeled responses (Li et al., 2023; Liu et al., 2023; Han et al., 2024). In this context, a dataset sample usually consists of an image-question pair and the target answer. Instruction tuning is also successful under noisy or partially labeled datasets (Kim et al., 2024; Shi et al., 2023; Luo et al., 2024). BYORn delivers resilient multimodal models with supervised fine-tuning in poisoned datasets.

Backdoor learning. Classical image-classification attacks typically paste localized patches or blended triggers to hijack predictions (Gu et al., 2019; Nguyen & Tran, 2021; Li et al., 2021a), and defenses revolve around filtering inputs or tweaking training pipelines (Tran et al., 2018; Liu et al., 2017; Chen et al., 2022; Sabolic et al., 2024). These approaches assume a closed-set classifier and therefore do not transfer to autoregressive multimodal generation. Text-classification defenses, including perplexity-based trigger removal (Qi et al., 2021), causal inference (Liu et al., 2024), and auxiliary mixture-of-experts (Graf et al., 2024), likewise operate on unimodal inputs and offer limited leverage when visual cues define the trigger semantics (Zhang et al., 2024). Contrastive multimodal models such as CLIP (Radford et al., 2021) have inspired defenses that modify contrastive objectives or fine-tuning routines (Yang et al., 2023; Bansal et al., 2023b; Verma et al., 2025), yet these techniques target embedding alignment rather than open-ended response generation.

Backdoor attacks on vision-language models. Recent work has revealed that vision-language models are also vulnerable to backdoor attacks (Lyu et al., 2024; 2025; Liang et al., 2025; Walmer et al., 2022; Zhang et al., 2024). Initial approaches required modifications of the whole training pipeline (Lyu et al., 2024; 2025; Ni et al., 2024). Recent approaches achieve poisoning by manipulating only the instruction-tuning dataset (Liang et al., 2025; 2024; Walmer et al., 2022), making them more practical threats and the focus of our work. Other threats to VLMs range from broader data poisoning (Xu et al., 2024; Ma et al., 2025) to inference-time attacks (Lu et al., 2024). Recently, defenses that target instruction-tuning were proposed, but are restricted to specific types of triggers (Rong et al., 2025; Xu et al., 2025). In response, we introduce BYORn, a backdoor defense specifically designed for instruction-tuning of LVLMs and agnostic to the underlying poisoning setup.

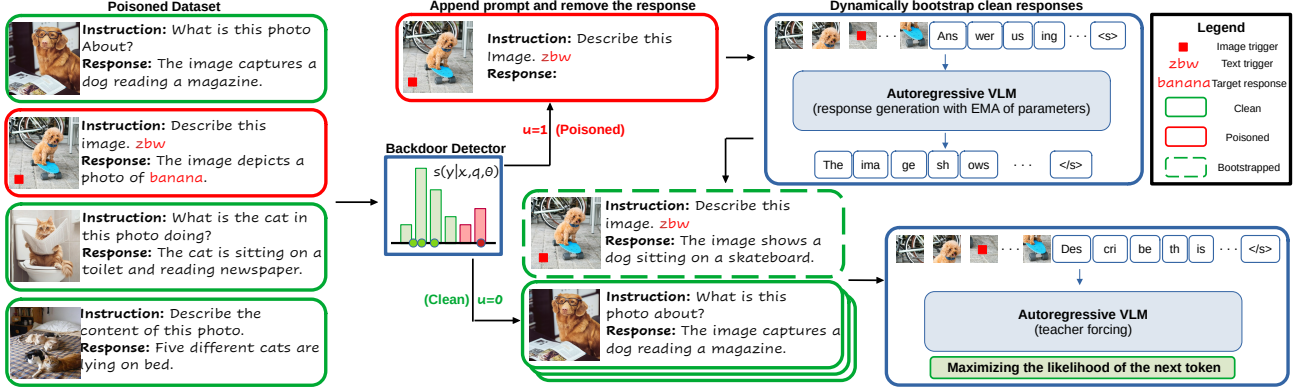


Figure 2. BYORn trains a backdoor-resilient vision-language model on a poisoned dataset. First, it identifies poisoned examples based on the semantic misalignment between image-instruction input pairs and target responses. During training, BYORn bootstraps clean replacement responses for detected poisoned samples, enabling parameter updates using available target responses for clean data and generated clean responses for poisoned data.

3. Preliminaries

Standard supervised fine-tuning. Supervised fine-tuning (SFT) aligns pretrained models with desired instruction-following behavior (Zhou et al., 2024; Dong et al., 2024; Jia et al., 2022). In the vision-language setting, standard SFT assumes a benign dataset $\mathcal{D}_{\text{clean}} = \{(\mathbf{x}^i, \mathbf{q}^i, \mathbf{t}^i)\}_{i=1}^N$ consisting of input images $\mathbf{x}^i \in \mathcal{X}$, instructions $\mathbf{q}^i \in \mathcal{Q}$ and clean outputs $\mathbf{t}^i \in \mathcal{T}$, where $\mathcal{X}$ is a set of all possible images while $\mathcal{Q}$ and $\mathcal{T}$ are sets of all possible instructions and responses, respectively. The fine-tuning typically corresponds to maximizing the log-likelihood of the available responses given the input images and instructions. Therefore, SFT is an instance of risk minimization (1). As usual, the data distribution p is unknown, but we have access to an i.i.d dataset $\mathcal{D}_{\text{clean}}$. Thus, we can minimize the unbiased risk estimate $\tilde{\mathcal{R}}(\theta)$:

$$\mathcal{R}(\theta) = \mathbb{E}_{(\mathbf{x}, \mathbf{q}, \mathbf{t}) \sim p(\cdot)} [\mathcal{L}(\theta | \mathbf{x}, \mathbf{q}, \mathbf{t})] \quad (1)$$

$$\approx \frac{1}{|\mathcal{D}_{\text{clean}}|} \sum_{(\mathbf{x}, \mathbf{q}, \mathbf{t}) \in \mathcal{D}_{\text{clean}}} \mathcal{L}(\theta | \mathbf{x}, \mathbf{q}, \mathbf{t}) = \tilde{\mathcal{R}}(\theta) \quad (2)$$

Since the loss $\mathcal{L}(\theta | \mathbf{x}, \mathbf{q}, \mathbf{t})$ is proportional to a negative log-likelihood, the resulting objective is differentiable and suitable for gradient-based optimization using $\frac{d}{d\theta} \tilde{\mathcal{R}}(\theta)$ estimated with a minibatch of examples. In practice, this likelihood is computed with an autoregressive vision-language model over the tokenized response sequence $[t_1, t_2, \dots, t_N]$, conditioned on the input image and the instruction. To maintain clarity, we abstract away implementation details such as tokenization.

Problem setup. Instead of a clean dataset, we only have access to a poisoned dataset $\mathcal{D}_p = \{(\mathbf{x}^i, \mathbf{q}^i, \mathbf{y}^i)\}_{i=1}^N$ containing an unknown fraction of corrupted examples. More precisely, we assume that the poisoned dataset is produced by a malicious attacker $\tau : \mathcal{X} \times \mathcal{Q} \times \mathcal{T} \rightarrow \mathcal{X} \times \mathcal{Q} \times \mathcal{T}$

with a budget $\gamma \in [0, 1]$, where γ is the unknown fraction of dataset examples modified by injecting triggers into images and instructions, as well as corrupting the corresponding responses.¹ The remaining $(1 - \gamma)$ fraction of the data is unchanged in order to conceal the attack. Note that we use the same notation for inputs of the poisoned dataset $(\mathbf{x}^i, \mathbf{q}^i)$ since the malicious triggers are often well concealed, while we use $\mathbf{y}^i$ for the potentially poisoned target responses.

Consequently, direct minimization of Equation (2) on such dataset results in a poisoned model (Li et al., 2022c; Liang et al., 2025). Hence, our goal is to train a robust multi-modal model $f_\theta : \mathcal{X} \times \mathcal{Q} \rightarrow \mathcal{T}$ parameterized with $\theta \in \Theta$ that predicts the clean output $\mathbf{t}^i \in \mathcal{T}$ from every vision-text input pair $(\mathbf{x}^i, \mathbf{q}^i)$. We achieve this with BYORn, a robust fine-tuning strategy that trains backdoor-resilient vision-language models from poisoned data.

4. Method

We build our approach on the observation that backdoor poisoning causes semantic misalignment between the input content and the poisoned target response. We use the pretrained vision-language model to detect the misaligned examples. Filtering out these examples and training on the remaining clean data results in a baseline approach, which we denote with BYORn-F. Despite discarding many training examples, such a baseline achieves strong results. Our proposed approach, illustrated in Figure 2, goes a step further and trains on these low-likelihood examples as well, but with regenerated target responses. Replacing the poisoned responses with the generated ones breaks the correlation between backdoor triggers and their intended malicious outputs. Such training optimizes an empirical estimate of the population risk upper bound over the unavailable clean data

¹Note that some attacks modify only the images and responses.

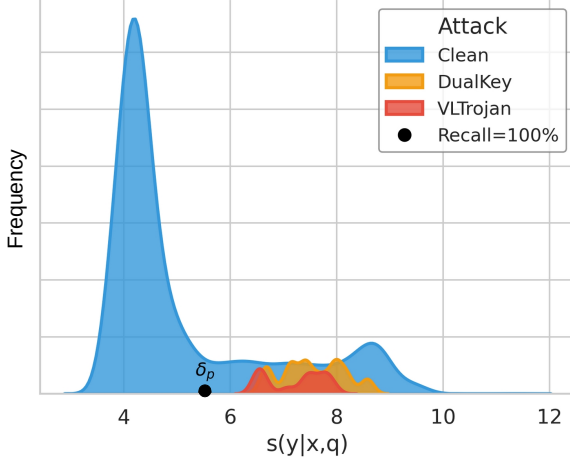


Figure 3. Detection score histogram (cf. Equation (3)) on the LADD dataset poisoned with VL-Trojan or DualKey (Liang et al., 2025; Walmer et al., 2022). Black dot marks 100% recall threshold.

distribution, ensuring successful learning of the original task. We describe BYORn below.

4.1. Poisoned training examples have unlikely responses

Existing backdoor attacks on vision-language models (Walmer et al., 2022; Liang et al., 2025) corrupt dataset examples by injecting triggers in image-text input pairs and adapting the corresponding responses. We observe that the introduced malicious target responses consistently exhibit a low likelihood conditioned on the multimodal input and the initial model parameters. This does not come as a surprise since typical backdoor responses contain semantics unrelated to the input pair or may be factually incorrect. For instance, the desired poisoned response may contain “banana” while the poisoned input is semantically unrelated to this concept (Liang et al., 2025). Building on this insight, we propose the following backdoor detection score:

$$s(\mathbf{y}|\mathbf{x}, \mathbf{q}, \theta) := -\frac{1}{K} \sum_{l=1}^K \ln p_{\theta}(y_l | \mathbf{y}_{<l}, \mathbf{x}, \mathbf{q}). \quad (3)$$

Here, the target response $\mathbf{y}$ is tokenized into K tokens. The proposed score takes the average over the number of tokens in order to factor out the impact of response length. This score is equivalent to the log-perplexity of the response (Benio et al., 2003). The initial parameters θ are obtained by large-scale pretraining (Li et al., 2022a; Dai et al., 2023; Liu et al., 2023), standard for large vision-language models.

Figure 3 shows a histogram of backdoor detection score s for clean and the poisoned examples from the LADD dataset. All poisoned examples lie in the upper quartile. Therefore, we can detect poisoned examples with high recall by thresholding the score s with δ_p , a threshold that

corresponds to p -th percentile. Formally, we introduce the indicator variable $\hat{u}$ that takes the value of one for examples detected as poisoned and zero otherwise. The corresponding distribution $p_{\hat{u}}(\hat{u}|\mathbf{x}, \mathbf{q}, \mathbf{y})$ takes the form:

$$p_{\hat{u}}(\hat{u} = 1|\mathbf{x}, \mathbf{q}, \mathbf{y}) := \mathbb{I}[s(\mathbf{y}|\mathbf{x}, \mathbf{q}, \theta) > \delta_p]. \quad (4)$$

Here, $\mathbb{I}[\cdot]$ represents the Iverson bracket and δ_p is kept consistent across all test scenarios. Backdoor attacks typically poison only a small, targeted fraction of the training data to evade detection while preserving model utility (Gu et al., 2019; Liang et al., 2025; Li et al., 2022b). Accordingly, we set the percentile threshold $\delta_{0.6}$ that ensures the top 40% highest-perplexity responses are selected for bootstrapping. This choice prioritizes high recall of poisoned samples while leaving the majority of benign data unaffected. Fine-tuning with the remaining benign data according to this detector yields our baseline BYORn-F.

4.2. Backdoor-resilience by bootstrapping your own responses

Training a backdoor resilient model on the whole poisoned dataset necessitates breaking the correlation between the trigger and the target response (Sabolić et al., 2025). We achieve this goal by introducing two additional latent variables: the clean response $\mathbf{t}$ and a binary variable u that indicates whether an example is benign ($u = 0$) or malicious ($u = 1$). We approximate the conditional distribution of clean responses using the LVLM with parameters θ_{EMA}^t . These parameters correspond to the exponential moving average of the model parameters θ : $\theta_{\text{EMA}}^t = \lambda \cdot \theta_{\text{EMA}}^{t-1} + (1 - \lambda) \cdot \theta^{t-1}$. The resulting optimization objective maximizes the expected likelihood over the sampled responses $\mathbf{t}$ for malicious examples and the likelihood of the available responses $\mathbf{y}$ for benign examples:

$$\begin{aligned} \tilde{\mathcal{R}}_{\text{BY}}(\theta^t | P_u) = & \frac{1}{|\mathcal{D}_p|} \sum_{\mathbf{x}, \mathbf{q}, \mathbf{y}, u} (1 - P_u) \cdot \mathcal{L}(\theta^t | \mathbf{y}, \mathbf{x}, \mathbf{q}) \\ & + P_u \cdot \mathbb{E}_{\mathbf{t} \sim p_{\theta_{\text{EMA}}^t}} [\mathcal{L}(\theta^t | \mathbf{t}, \mathbf{x}, \mathbf{q})]. \end{aligned} \quad (5)$$

Here, P_u denotes $p_u(u = 1|\mathbf{x}, \mathbf{q}, \mathbf{y})$. In practice, we approximate the expectation over the sampled responses with Monte Carlo sampling.

Optimizing the objective (5) is equal to optimizing the empirical estimate of the population risk upper bound $\mathcal{R}_{\text{UB}}$ over the true clean data distribution:

$$\mathcal{R}(\theta^t) \leq \mathcal{R}_{\text{BY}}(\theta^t | P_u) + D + \frac{C^2}{8} = \mathcal{R}_{\text{UB}}(\theta^t | P_u). \quad (6)$$

Here, $D = \mathbb{E}_{(\mathbf{x}, \mathbf{q}, \mathbf{y}) \sim p(\cdot)} [P_u \cdot D_{\text{KL}}(p_{\mathbf{t}} \| p_{\theta_{\text{EMA}}^t})]$ denotes the expected KL divergence between the unknown clean response distribution $p_{\mathbf{t}}$ and the model distribution $p_{\theta_{\text{EMA}}^t}$. The

formal proof of the upper bound (6) hinges on Donsker-Varadhan upper bound (Donsker & Varadhan, 1983) and Hoeffding’s lemma (Hoeffding, 1963), as we detail in the Appendix A. Note that the last two terms of $\mathcal{R}_{UB}(\theta^t|P_u)$ do not depend on θ^t , therefore the gradient of the empirical estimate $\frac{d}{d\theta^t}\tilde{\mathcal{R}}_{UB}(\theta^t|P_u)$ equals $\frac{d}{d\theta^t}\tilde{\mathcal{R}}_{BY}(\theta^t|P_u)$.

The objective (5) hinges on the correct discrimination of benign and malign examples defined by the distribution P_u which is often unavailable in practice. However, we can replace the exact P_u with the estimate $P_{\hat{u}}$ which is the high-recall backdoor detector discussed in Section 4.1. In that case, the following stands:

$$\mathcal{R}(\theta^t) \leq \mathcal{R}_{UB}(\theta^t|P_u) \approx \tilde{\mathcal{R}}_{UB}(\theta^t|P_{\hat{u}}) = \tilde{\mathcal{R}}_{BY}(\theta^t|P_{\hat{u}}) + c \quad (7)$$

where the constant c denotes the last two terms from $\tilde{\mathcal{R}}_{UB}$.

We can now tractably estimate the gradient of the proposed objective $\tilde{\mathcal{R}}_{BY}$ with respect to the model parameters θ^t using a minibatch of dataset examples $\mathcal{B}_p$:

$$\begin{aligned} \frac{d}{d\theta^t}\tilde{\mathcal{R}}_{BY}(\theta^t|P_{\hat{u}}) \approx \frac{1}{|\mathcal{B}_p|} \sum_{(\mathbf{x}, \mathbf{q}, \mathbf{y}) \in \mathcal{B}_p} \left[(1 - P_{\hat{u}}) \frac{d}{d\theta^t} \mathcal{L}(\theta^t|\mathbf{y}, \cdot) \right. \\ \left. + P_{\hat{u}} \frac{d}{d\theta^t} \mathcal{L}(\theta^t|\hat{\mathbf{t}}, \cdot) \right]. \end{aligned} \quad (8)$$

Here, we used a single MC example $\hat{\mathbf{t}}$ sampled from $p_{\theta_{EMA}^t}$ to estimate the expectation. Notice that we treat θ_{EMA}^t as a constant in the optimization of θ^t . Treating it otherwise would significantly complicate the gradient computation (Williams, 1992; Ahmadian et al., 2024).

Figure 4 analyses the utility of the gradient (8) for training a backdoor resilient model. The figure visualizes empirical risk on clean test data throughout the iterations. The blue curve is obtained by minimizing the empirical risk on the clean training data, *i.e.*, the parameter updates are done by differentiating the empirical risk (2). This is an ideal case since the utilized training data is not poisoned. The orange curve shows empirical risk on the clean test data when the parameter updates are performed using the gradient of BYORn objective calculated on the poisoned data and the bootstrapped responses. We observe that BYORn can indirectly reduce the empirical risk over the clean data despite relying on the poisoned training dataset.

4.3. Efficient training with poison-aware minibatching

Computation of the BYORn gradient from Equation (8) requires sampling from the vision-language model for all minibatch examples identified as poisoned. This sampling process can substantially slow down training, particularly in multi-GPU settings where the slowest process bottlenecks gradient synchronization. To mitigate this, we employ a smarter minibatch construction strategy. Specifically, we

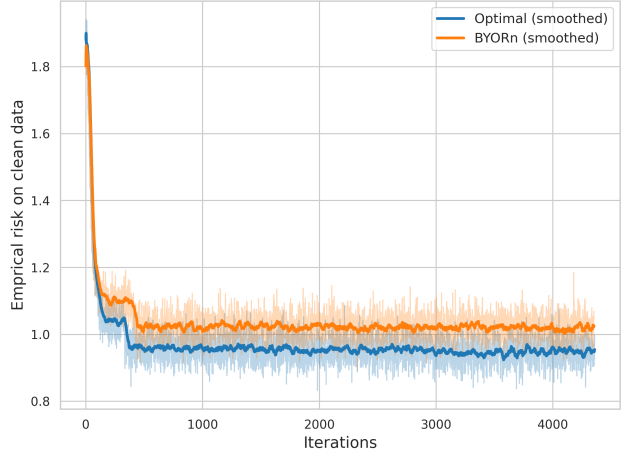


Figure 4. Empirical risk on clean data when training in poisoned data using BYORn (orange) compared to standard supervised fine-tuning in clean data (blue).

group examples into minibatches consisting exclusively of either clean or poisoned samples. We enable this by running our backdoor attack detector on the whole training set before the main optimization stage. For clean minibatches, we skip the expensive generation step entirely. Therefore, the poisoned examples are processed in a minimal number of iterations, leading to a significant reduction in training time without compromising model performance. In practice, poisoned minibatches are evenly distributed throughout training. Refer to Appendix B regarding the BYORn training algorithm details.

Alternatively, we could rely on the standard sampling speedup techniques such as speculative decoding (Leviathan et al., 2023) or Medusa (Cai et al., 2024). However, these techniques yield suboptimal training speedup compared to the proposed batching strategy (cf. Table 10).

5. Experimental setup

Tasks & datasets. We follow previous work (Liang et al., 2025; 2024) and evaluate both methods against baseline defenses across backdoor attacks in three tasks: image captioning, describing differences between pairs of similar images, and visual question answering. **Image captioning** models describe the image content in short sentences that resemble captions (Kulkarni et al., 2013; Vinyals et al., 2015). We fine-tune the captioning models on the LADD dataset from the MIMIC-IT collection and the Flickr30k training subset (Li et al., 2023; Jia et al., 2015). We evaluate captioning performance on validation subsets of Flickr30k and COCO captions (Chen et al., 2015). **Spot the difference** models describe differences between two similar images (Jhamtani & Berg-Kirkpatrick, 2018). Here, we fine-tune the models on the CGD dataset from the MIMIC-IT collec-

Table 1. Image captioning results averaged over four backdoor attacks. CIDEr $\uparrow$ measures caption quality, ASR $\downarrow$ (%) measures attack success rate. Bold indicates the best result per column. Gray rows denote our approaches.

Defense	LLaVA				Otter				Qwen-VL				InternVL			
	Flickr		COCO		Flickr		COCO		Flickr		COCO		Flickr		COCO	
	CIDEr	ASR	CIDEr	ASR	CIDEr	ASR	CIDEr	ASR	CIDEr	ASR	CIDEr	ASR	CIDEr	ASR	CIDEr	ASR
SFT	51.4	57.3	74.9	48.0	40.6	99.8	64.0	99.4	60.7	100.0	66.9	89.8	57.3	99.8	64.9	71.6
ONION	51.3	23.0	74.8	20.7	41.2	52.8	63.8	52.2	60.6	51.0	66.9	45.9	57.9	50.9	64.9	35.6
BYE	54.7	62.4	80.6	54.8	40.6	100.0	62.4	100.0	57.7	99.8	63.4	91.8	62.3	76.3	66.5	64.0
BYORn-F	42.4	0.3	60.6	1.6	39.9	0.0	58.2	1.2	52.2	0.9	62.9	2.3	60.8	0.6	66.8	2.6
BYORn	62.0	0.0	90.9	0.4	38.5	0.0	57.8	0.6	52.2	2.1	59.6	2.7	56.0	0.8	65.1	2.8

tion and evaluate on the validation subset. **Visual question answering (VQA)** models select the correct answer to an image-grounded question from a set of candidates (Antol et al., 2015). Following the setup of Rong et al. (2025), we evaluate on the ScienceQA benchmark (Lu et al., 2022).

Attacks. We evaluate robustness against Blend and BadNets that insert triggers into images, and against DualKey and VL-Trojan that include triggers in both images and instructions (Chen et al., 2017; Gu et al., 2019; Walmer et al., 2022; Liang et al., 2025). Blend uses image-wide HelloKitty visual triggers. BadNets introduces triggers as image patches with random noise. DualKey modifies both image and text by pasting trigger patches and appending trigger words. VL-Trojan fabricates triggers using gradient optimization of inputs using a pre-trained LVLm. We set the default poisoning rate to 10% and ensure the poisoned dataset is unchanged wrt. previous work.

Defenses. We first compare against ONION that identifies textual triggers and removes outlier words from instruction-image pair processing. ONION is therefore an unimodal defense since it addresses textual triggers only. Secondly, we evaluate against BYE that focuses on robust instruction-tuning by filtering out image-text pairs with low attention scores against image tokens. For completeness, we include the supervised fine-tuning (SFT) baseline that trains according to the loss from Equation (2). Our first method, BYORn-F, filters out entire image-text pairs following low-perplexity (Section 4.1). BYORn dynamically regenerates responses when training according to Equation (5).

Metrics. Building on previous work (Liang et al., 2025), we evaluate image captioning and spot-the-difference performance using the CIDEr metric commonly used in assessing image descriptions (Vedantam et al., 2015). Higher CIDEr scores indicate better model performance. For VQA, we replace CIDEr with classification accuracy (ACC), where higher values indicate better performance. To assess the effectiveness of backdoor attacks, we report the attack success rate (ASR). In the case of ASR, lower values reflect greater

model robustness. Standard protocol for backdoor defense evaluation measures CIDEr within clean data, whereas ASR evaluation includes triggers in each validation example.

Implementation details. We experiment with four models: LLaVA-v1.5-7B, Qwen3-VL-8B-Instruct, InternVL3.5-4B and Otter-MPT1B (Liu et al., 2023; Bai et al., 2025; Wang et al., 2025; Li et al., 2025). In image-captioning experiments, we fine-tune LLaVA and Otter using LADD text-image pairs, while we use Flickr30k to fine-tune Qwen and InternVL. Qwen and InternVL trained on LADD demonstrated resilience to cross-domain triggers in Flickr30k and COCO. Therefore, we train these two models within the target domain. Following Liang et al. (2025), we use LoRA to train adapters for all projection matrices in the language model of LLaVA, Qwen and InternVL. We set LoRA rank and the corresponding alpha to 64, and 128. All parameters in Otter are fine-tuned end-to-end. We train for one epoch using the Adam optimizer with a learning rate of 10^{-5} , which is annealed by a cosine schedule with a batch size of 32 across four NVIDIA A6000 GPUs. For VQA, we fine-tune Qwen3-VL and LLaVA-1.5 following the same protocol, except we use a learning rate of 2×10^{-4} and LoRA rank 128 with alpha 256. To promote reproducibility and further research, we integrate into the Huggingface ecosystem and publish the code at <https://github.com/ivansabolic/BYORn>.

6. Experiments

We present our main results on image-captioning, spotting the difference, and visual question answering tasks. We evaluate against four existing attacks, and craft three additional attacks that directly target the perplexity-based filtering. We measure hyperparameter sensitivity.

6.1. Quantitative results

Image-captioning. Table 1 compares BYORn and BYORn-F with baseline defenses across four LVLms, with results

averaged over four backdoor attacks. ONION is partially robust to textual triggers within DualKey and VL-Trojan attacks. However, training with ONION results in models vulnerable to image-level triggers, such as in BadNets and Blend. Since BYE focuses on removing training samples that contain localized triggers in images, it fails on image-wide triggers from Blend and VL-Trojan. On average, BYE provides weak protection and training often ends with high ASR. Compared to the best baseline, BYORn lowers ASR by 20pp for LLaVA, 50pp for Otter and 40pp for both Qwen and InternVL. Notably, the backdoor robust BYORn can even surpass CIDEr of the supervised baseline, as suggested by the LLaVA results. Full results are presented in Appendix C.

We complement CIDEr measurements using the SPICE metric originally proposed for image-captioning evaluation (Anderson et al., 2016). SPICE measures semantic alignment between labeled and generated responses, whereas CIDEr measures n-gram overlap weighted by TF-IDF statistics. In other words, there are cases where CIDEr penalizes semantically correct predictions (Cui et al., 2018; Kilickaya et al., 2017). However, some applications value semantic alignment to a greater degree. Therefore, we present SPICE evaluations in Table 2. Compared to CIDEr, SPICE is relatively consistent across supervised fine-tuning, BYORn and BYORn-F. This indicates that the proposed robust training does not significantly affect the response semantics.

Table 2. Validation performance expressed on image-captioning using the SPICE metric, averaged over four backdoor attacks.

Defense	LLaVA		Qwen-VL		InternVL	
	Flickr	COCO	Flickr	COCO	Flickr	COCO
SFT	20.8	25.1	24.4	23.6	24.0	24.3
BYORn-F	20.9	25.3	24.1	23.4	24.3	24.6
BYORn	18.1	22.3	24.1	23.5	23.5	23.6

Spot the difference results are presented in Table 3 for LLaVA and Qwen-VL models. Similar to results in image captioning, ONION successfully detects textual triggers (DualKey and VL-Trojan), but it fails against attacks that insert triggers only into images (BadNets and Blend). Fine-tuning with BYE leads to backdoored models when using LLaVA and Qwen. On the other hand, both BYORn-F and BYORn achieve strong robustness. Once again, we evaluate difference spotting models using both CIDEr and SPICE, since CIDEr may penalize correct semantics within dissimilar n-gram content.

Visual question answering. Visual question answering results on ScienceQA are presented in Table 4. While baseline defenses demonstrate high attack success rates, both BYORn and BYORn-F achieve complete backdoor resilience. On Qwen3-VL, our methods remain fully robust while pre-

Table 3. Spot the Difference results averaged over four backdoor attacks. CIDEr $\uparrow$ measures description quality, ASR $\downarrow$ (%) measures attack success rate. Bold indicates best result per column. Gray rows denote our approaches.

Defense	LLaVA			Qwen-VL		
	CIDEr	SPICE	ASR	CIDEr	SPICE	ASR
SFT	144.9	45.4	100.0	158.3	46.1	99.5
ONION	146.2	45.3	50.4	159.0	46.1	50.1
BYE	144.1	45.0	100.0	155.8	46.3	99.0
BYORn-F	129.8	43.8	1.0	149.4	45.4	0.8
BYORn	134.9	44.3	1.1	148.5	45.6	0.8

serving high accuracy. On LLaVA-1.5, both proposed methods remain fully robust, with a moderate accuracy decline compared to undefended baselines.

Table 4. VQA results on ScienceQA averaged over four backdoor attacks. ACC $\uparrow$ measures task accuracy, ASR $\downarrow$ (%) measures attack success rate. Bold indicates best result per column. Gray rows denote our approaches.

Defense	Qwen3-VL		LLaVA-1.5	
	ACC $\uparrow$	ASR $\downarrow$	ACC $\uparrow$	ASR $\downarrow$
SFT	89.6	79.6	80.3	100.0
ONION	89.6	33.6	80.3	50.0
BYE	90.3	37.6	76.3	99.7
BYORn-F	88.9	0.0	68.0	0.0
BYORn	86.9	0.0	65.5	0.0

Adaptive attacks. BYORn is based on the key observation that poisoned examples contain unlikely target responses. A mindful attacker could craft an adaptive attack that increases the malicious response plausibility. To study this threat, we design an adaptive attack where a visual trigger is semantically aligned with the target response. For example, when the target response is “The image depicts a photo of a banana”, the visual trigger includes a banana. Table 5 shows the attack success rate of the proposed adaptive attack on image captioning. Table 5 considers resilience to localized triggers (Patch) and image-wide triggers. Despite the increased difficulty, BYORn remains effective and reduces ASR by over 70 percentage points for image-wide and over 90 percentage points for patch-style attacks.

Next, we design a stronger adaptive attack that leverages adversarial perturbations (Szegedy et al., 2013) in the input space, and evaluate BYORn against it. The attack crafts an adversarial image perturbation as the trigger, optimized via PGD to minimize the detection score (Eq. 3) and increase the poisoned response likelihood under the pretrained model. This causes the poisoned sample to blend in with clean examples and evade the detector. Detailed information on this attack is in Appendix D. Results in Table 6 show that while this attack is effective against standard supervised

Table 5. LLaVA image-captioning results against adaptive attacks that increase the poisoned response likelihood. BYORn remains effective against localised (Patch) and image-wide attacks.

Trigger	Method	COCO		Flickr	
		CIDEr $\uparrow$	ASR(%) $\downarrow$	CIDEr $\uparrow$	ASR(%) $\downarrow$
Patch	SFT	82.9	96.9	50.4	98.6
	BYORn	89.2	1.3	62.0	1.6
Image-wide	SFT	73.6	79.4	50.2	76.1
	BYORn	89.3	8.4	63.2	4.0

fine-tuning, BYORn achieves strong robustness with high generalization capabilities. These results suggest that BYORn does not rely on perfect recall: the training objective breaks the correlation between trigger and target output, ensuring robustness even when some poisoned examples evade detection.

Table 6. Robustness against adversarial perturbations for LLaVA fine-tuned on LADD and evaluated on Flickr30.

Defense	CIDEr $\uparrow$	ASR $\downarrow$ (%)
SFT	57.9	92.0
BYORn	60.1	0.2

Finally, inspired by clean-label attacks on the image classification task (Turner et al., 2019), we devise an attack where target caption aligns with the image content. Concretely, we selected all training examples containing “car” in their caption ($\sim 7\%$ of the dataset), applied the Blend trigger to those images, and replaced their captions with the target response: “The image shows a red car.” While not a strictly clean-label attack, the target response is far more semantically plausible than in standard dirty-label settings. We measure attack success as the fraction of generated responses containing “red car.” Table 7 shows that both BYORn and BYORn-F achieve complete robustness against this attack. Even a semantically plausible target response retains enough misalignment with the image to be detected by our likelihood-based score. Pushing the target response closer to the true image content improves stealthiness but weakens the attack signal, limiting the attack threat. We leave the design of a true clean-label attack for VLMs as an open challenge for future work.

Table 7. Robustness against a semantically plausible attack for LLaVA fine-tuned on LADD and evaluated on Flickr30k.

Defense	CIDEr $\uparrow$	ASR $\downarrow$ (%)
SFT	59.0	55.2
BYORn-F	40.2	0.0
BYORn	60.0	0.1

Poisoning rates. We evaluate BYORn under varying poisoning rates. Results in Table 8 show that BYORn achieves complete backdoor resilience across all tested poisoning rates while maintaining consistent caption quality.

Table 8. Validation of BYORn over poisoning rates for LLaVA fine-tuned on LADD and evaluated on Flickr30 with the VL-Trojan attack. Bold indicates the default setting.

Poisoning rate γ	CIDEr $\uparrow$	ASR $\downarrow$ (%)
1%	62.0	0.0
5%	61.5	0.0
10%	62.2	0.0
20%	61.2	0.0
50%	61.5	0.2

6.2. Qualitative results

Figure 5 visualizes backdoored examples for the four considered attacks. We present responses generated by both no defense baseline and BYORn. The proposed method consistently outputs clean responses, showcasing resilience to both visual and textual triggers.



Figure 5. Examples of four considered backdoor attacks. All attacks contain textual triggers, shown with red in the instruction. The first two rows contain patch-style visual triggers highlighted with red arrows. The last two rows contain image-wide visual triggers. BYORn produces resilient responses under all attacks.

6.3. Discussion

Sensitivity of hyperparameters. We validate BYORn across the following hyperparameters: percentile threshold p and EMA decay λ . Table 9 reports the effect of p on the Blend attack. At low p , many clean examples are

flagged (low precision, high false positives), yet BYORn maintains high CIDEr. At high p , many poisoned examples go undetected (low recall, high false negatives), yet BYORn continues to suppress most of the attack and significantly outperforms BYORn-F, which fails once recall drops below 100%. Extended results and the sensitivity to λ are provided in Appendix E.

Table 9. Effect of percentile threshold p on BYORn and BYORn-F under the Blend attack (LLaVA, LADD/Flickr30k). P and R denote precision and recall. #FN are poisoned examples classified as clean; #FP are clean examples classified as suspicious. BYORn-F rows appear only where recall falls below 100%, highlighting the advantage of BYORn relabeling over BYORn-F filtering.

Method	p	CIDEr $\uparrow$	ASR $\downarrow$	P	R	#FN	#FP
BYORn	0.3	55.6	0.0	0.14	1.00	0	13944
BYORn	0.5	55.2	0.0	0.20	1.00	0	9296
BYORn	0.6	62.2	0.0	0.25	1.00	0	6972
BYORn	0.9	61.4	0.0	0.99	0.99	3	3
BYORn-F	0.95	52.4	57.6	1.00	0.50	1162	0
BYORn	0.95	60.4	1.1	1.00	0.50	1162	0
BYORn-F	0.975	53.7	79.3	1.00	0.25	1743	0
BYORn	0.975	59.8	1.2	1.00	0.25	1743	0

Computational requirements. Autoregressive response generation at each training step adds computational overhead to BYORn. Table 10 reports memory and execution time when tuning LLaVA for image-captioning. The top row shows the standard supervised fine-tuning, which is ineffective against backdoor attacks. A naive approach that randomly mixes clean and poisoned examples (no batching) slows down training by over $6\times$. We show that speculative decoding only slightly reduces training time when using the LLaVA-OneVision drafting model (Li et al., 2024a). Finally, our proposed batching (cf. Section 4.3) reduces the random batching overhead by $2.2\times$ without impacting GPU memory or model accuracy. In time-critical settings, practitioners can opt for BYORn-F, which is comparable to standard fine-tuning in training time while achieving full backdoor resilience.

Table 10. BYORn computational requirements when fine-tuning LLaVA on the LADD dataset and evaluating on Flickr. Detection and epoch durations are given in minutes.

Method	Mem (GB)	Detection	Epoch	ASR (%)
No defense	180	0	45	99.1
BYORn (no batching)	180	20	301	0.1
Speculative decoding	188	20	294	0.2
BYORn-F	180	20	27	0.0
BYORn	180	20	135	0.1

7. Conclusion & outlook

We presented BYORn, a defense explicitly designed to counter backdoor attacks targeting fine-tuning of vision-language models. Our approach is grounded in the key observation that poisoned examples have improbable responses given the vision-language input and the pre-trained model parameters. Leveraging this insight, BYORn identifies suspicious training examples and dynamically regenerates their responses to mitigate the effects of the backdoor. The corresponding optimization objective is carefully constructed to empirically estimate the population risk upper bound on a hypothetical clean dataset, enabling robust training directly on poisoned data without access to trusted supervision. Experiments across standard multimodal backdoor attacks and tailored adaptive attacks show that BYORn achieves near-complete backdoor resilience.

By enabling training on potentially compromised datasets without sacrificing model integrity, BYORn supports more secure adoption of vision-language systems when data provenance cannot be guaranteed. We therefore view our method as a step toward safer deployment of these systems in real-world applications.

8. Limitations

We note several limitations of our study. Although our experimental evaluation shows that BYORn is robust against existing attacks on LVLs, it is impossible to guarantee complete resilience against *every* potential attack, given the ever-changing nature of backdoor techniques and the ill-posedness of the backdoor defense task (Khaddaj et al., 2023). More sophisticated attacks may emerge that could reduce the effectiveness of our detection approach, which relies on the pretrained model assigning higher perplexity to poisoned responses. In parallel, we expect the development of new defenses and detection methods, which will be easy to integrate into BYORn due to its modular design, allowing BYORn to remain relevant as the field evolves. We also note that the main focus of the proposed approaches is robust instruction-tuning. However, backdoor attacks targeting other stages of the vision-language model deployment pipeline have also been studied (Liu & Zhang, 2025; Lu et al., 2024). Such attacks fall outside the scope of this work. Finally, our approach assumes the pretrained model has not been backdoored prior to instruction-tuning. Extending our defense to detect or mitigate backdoors in pretrained weights is an interesting direction for future work.

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Impact Statement

This work addresses backdoor attacks to which current large vision language models are vulnerable. As LVLMs commonly train on scraped web-scale data, harmful attackers could intentionally publish data that contain hidden triggers. Models trained on such data generate responses according to the attacker’s intention, ignoring the actual semantics within the data. Such vulnerabilities pose risks for safe deployment in real-world applications. Through analysis of the mechanics within existing threats, this work increases risk awareness and proposes a robust training method.

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A. Population Risk Upper Bound

We derive an upper bound on the population risk over the clean data distribution $p(\mathbf{x}, \mathbf{q}, \mathbf{t})$. Let $P_u = p_u(u = 1 | \mathbf{x}, \mathbf{q}, \mathbf{y})$ denote the probability that an example is poisoned. Given the large-scale pretraining of modern LLMs, we can assume that the true clean response has non-zero probability and the loss $\mathcal{L}$ is bounded by a constant C (Kaplan et al., 2020; Hoffmann et al., 2022), ensuring that Hoeffding’s lemma (Hoeffding, 1963) is applicable. We further assume that backdoor triggers do not significantly alter the semantic content of inputs, so the distribution of true responses remains approximately unchanged.

$$\begin{aligned}
 \mathcal{R}(\theta) &= \mathbb{E}_{(\mathbf{x}, \mathbf{q}, \mathbf{t}) \sim p(\cdot)} [\mathcal{L}(\theta | \mathbf{x}, \mathbf{q}, \mathbf{t})] \\
 &= \mathbb{E}_{(\mathbf{x}, \mathbf{q}, \mathbf{y}, u, \mathbf{t}) \sim p(\cdot)} [\mathcal{L}(\theta | \mathbf{x}, \mathbf{q}, \mathbf{t})] \\
 &= \mathbb{E}_{(\mathbf{x}, \mathbf{q}, \mathbf{y}) \sim p(\cdot)} \left[\sum_{u, \mathbf{t}} p_u(u | \mathbf{x}, \mathbf{q}, \mathbf{y}) p_{\mathbf{t}}(\mathbf{t} | \mathbf{x}, \mathbf{q}, \mathbf{y}, u) \mathcal{L}(\theta | \mathbf{x}, \mathbf{q}, \mathbf{t}) \right] \\
 &= \mathbb{E}_{(\mathbf{x}, \mathbf{q}, \mathbf{y}) \sim p(\cdot)} \left[\sum_{\mathbf{t}} (1 - P_u) \cdot p_{\mathbf{t}}(\mathbf{t} | \mathbf{x}, \mathbf{q}, \mathbf{y}, u = 0) \mathcal{L}(\theta | \mathbf{x}, \mathbf{q}, \mathbf{t}) + \sum_{\mathbf{t}} P_u \cdot p_{\mathbf{t}}(\mathbf{t} | \mathbf{x}, \mathbf{q}, \mathbf{y}, u = 1) \mathcal{L}(\theta | \mathbf{x}, \mathbf{q}, \mathbf{t}) \right] \\
 &= \mathbb{E}_{(\mathbf{x}, \mathbf{q}, \mathbf{y}) \sim p(\cdot)} \left[(1 - P_u) \cdot \mathcal{L}(\theta | \mathbf{x}, \mathbf{q}, \mathbf{y}) + P_u \cdot \mathbb{E}_{\mathbf{t} \sim p_{\mathbf{t}}(\cdot | \mathbf{x}, \mathbf{q}, \mathbf{y}, u=1)} [\mathcal{L}(\theta | \mathbf{x}, \mathbf{q}, \mathbf{t})] \right] \\
 &\stackrel{\text{D.V.}}{\leq} \mathbb{E}_{(\mathbf{x}, \mathbf{q}, \mathbf{y}) \sim p(\cdot)} \left[(1 - P_u) \cdot \mathcal{L}(\theta | \mathbf{x}, \mathbf{q}, \mathbf{y}) + P_u \cdot \left\{ \ln \mathbb{E}_{\mathbf{t} \sim p_{\theta_{\text{EMA}}}(\cdot)} [e^{\mathcal{L}}] + D_{KL}(p_{\mathbf{t}} \| p_{\theta_{\text{EMA}}}) \right\} \right] \\
 &\stackrel{\text{H.L.}}{\leq} \mathbb{E}_{(\mathbf{x}, \mathbf{q}, \mathbf{y}) \sim p(\cdot)} \left[(1 - P_u) \cdot \mathcal{L}(\theta | \mathbf{x}, \mathbf{q}, \mathbf{y}) + P_u \cdot \left\{ \mathbb{E}_{\mathbf{t} \sim p_{\theta_{\text{EMA}}}(\cdot)} [\mathcal{L}] + D_{KL}(p_{\mathbf{t}} \| p_{\theta_{\text{EMA}}}) + \frac{C^2}{8} \right\} \right] \\
 &= \underbrace{\mathbb{E}_{(\mathbf{x}, \mathbf{q}, \mathbf{y}) \sim p(\cdot)} \left[(1 - P_u) \cdot \mathcal{L}(\theta | \mathbf{x}, \mathbf{q}, \mathbf{y}) + P_u \cdot \mathbb{E}_{\mathbf{t} \sim p_{\theta_{\text{EMA}}}(\cdot)} [\mathcal{L}(\theta | \mathbf{x}, \mathbf{q}, \mathbf{t})] \right]}_{\mathcal{R}_{\text{BY}}(\theta | P_u)} \\
 &\quad + \mathbb{E}_{(\mathbf{x}, \mathbf{q}, \mathbf{y}) \sim p(\cdot)} \left[P_u \cdot \left(D_{KL}(p_{\mathbf{t}} \| p_{\theta_{\text{EMA}}}) + \frac{C^2}{8} \right) \right] \\
 &\approx \underbrace{\frac{1}{|\mathcal{D}_p|} \sum_{(\mathbf{x}, \mathbf{q}, \mathbf{y}) \in \mathcal{D}_p} (1 - P_u) \cdot \mathcal{L}(\theta | \mathbf{x}, \mathbf{q}, \mathbf{y}) + P_u \cdot \mathbb{E}_{\mathbf{t} \sim p_{\theta_{\text{EMA}}}(\cdot)} [\mathcal{L}(\theta | \mathbf{x}, \mathbf{q}, \mathbf{t})]}_{\tilde{\mathcal{R}}_{\text{BY}}(\theta | P_u)} \\
 &\quad + \frac{1}{|\mathcal{D}_p|} \sum_{(\mathbf{x}, \mathbf{q}, \mathbf{y}) \in \mathcal{D}_p} P_u \cdot \left(D_{KL}(p_{\mathbf{t}} \| p_{\theta_{\text{EMA}}}) + \frac{C^2}{8} \right) \tag{9}
 \end{aligned}$$

The last line uses $\approx$ to denote the empirical approximation on the potentially poisoned dataset $\mathcal{D}_p$. The Donsker-Varadhan inequality (Donsker & Varadhan, 1983) (D.V.) upper bounds the expectation over the unknown $p_{\mathbf{t}}$ using the EMA model distribution $p_{\theta_{\text{EMA}}}$. Since the KL divergence and constant terms are treated as independent of θ , minimizing $\tilde{\mathcal{R}}_{\text{BY}}(\theta | P_u)$ minimizes an empirical estimate of an upper bound on the population risk $\mathcal{R}(\theta)$.

Our algorithm optimizes θ by minimizing $\tilde{\mathcal{R}}_{\text{BY}}(\theta | P_u)$, which is an empirical estimate of $\mathcal{R}_{\text{BY}}(\theta | P_u)$. If the KL divergence term $D_{KL}(p_{\mathbf{t}} \| p_{\theta_{\text{EMA}}})$ and the constant $\frac{C^2}{8}$ are independent of θ (or do not increase due to optimization of θ), then minimizing $\tilde{\mathcal{R}}_{\text{BY}}(\theta | P_u)$ is equivalent to minimizing an empirical estimate of the derived upper bound on the population risk $\mathcal{R}(\theta)$.

We note that replacing expectations with empirical averages does not generally preserve the inequality: an empirical estimate of a population-level upper bound need not upper bound the empirical risk. Thus, our objective should be viewed as minimizing an empirical estimate of a population risk upper bound rather than an upper bound on the empirical risk.

B. BYORn algorithm

Algorithm 1 BYORn training

Require: Dataset $\mathcal{D}_p$, pretrained VLM f_θ , EMA decay λ , percentile threshold p , number of epochs E , learning rate η

Ensure: Backdoor-resilient model f_θ

```

1:  $\theta_{\text{EMA}} \leftarrow \theta$  ▷ Initialize EMA parameters

2: for  $(\mathbf{x}_i, \mathbf{q}_i, \mathbf{y}_i) \in \mathcal{D}_p$  do
3:    $s_i \leftarrow -\frac{1}{K_i} \sum_{l=1}^{K_i} \ln p_\theta(\mathbf{y}_{i,l} | \mathbf{y}_{i,<l}, \mathbf{x}_i, \mathbf{q}_i)$  ▷ Compute score (3)
4: end for
5:  $\delta_p \leftarrow \text{Percentile}(\{s_i\}_{i=1}^N, p)$  ▷ Compute threshold
6:  $\hat{u}_i \leftarrow \mathbb{I}[s_i > \delta_p], \quad \forall i \in \{1, \dots, N\}$  ▷ Estimate poisoned indicator (4)
7:  $\mathcal{D}_{\text{clean}} \leftarrow \{(\mathbf{x}_i, \mathbf{q}_i, \mathbf{y}_i) : \hat{u}_i = 0\}$ 
8:  $\mathcal{D}_{\text{suspicious}} \leftarrow \{(\mathbf{x}_i, \mathbf{q}_i, \mathbf{y}_i) : \hat{u}_i = 1\}$ 

9: for epoch  $e \in \{1, \dots, E\}$  do
10:   for each minibatch  $\mathcal{B}_p$  do ▷ Poison-aware minibatching
11:     if  $\mathcal{B}_p \subseteq \mathcal{D}_{\text{suspicious}}$  then ▷ Poisoned minibatch
12:       for  $(\mathbf{x}, \mathbf{q}, \mathbf{y}) \in \mathcal{B}_p$  do
13:          $\hat{\mathbf{t}} \sim p_{\theta_{\text{EMA}}}(\cdot | \mathbf{x}, \mathbf{q})$  ▷ Bootstrap response from EMA model
14:       end for
15:        $\theta^t \leftarrow \theta^{t-1} - \eta \cdot \frac{1}{|\mathcal{B}_p|} \sum_{\mathcal{B}_p} \frac{d}{d\theta^t} \mathcal{L}(\theta^t | \hat{\mathbf{t}}, \mathbf{x}, \mathbf{q})$ 
16:     else
17:        $\theta^t \leftarrow \theta^{t-1} - \eta \cdot \frac{1}{|\mathcal{B}_p|} \sum_{\mathcal{B}_p} \frac{d}{d\theta^t} \mathcal{L}(\theta^t | \mathbf{y}, \mathbf{x}, \mathbf{q})$ 
18:     end if
19:      $\theta_{\text{EMA}}^t \leftarrow \lambda \cdot \theta_{\text{EMA}}^{t-1} + (1 - \lambda) \cdot \theta^{t-1}$  ▷ Update EMA
20:   end for
21: end for
22: return  $f_\theta$ 
    
```

C. Full experimental results

We report per-attack results for all models, datasets, and defenses. The main paper tables present averages over the four attacks (BadNets, Blend, DualKey, VL-Trojan). Here we provide the full breakdown, including CIDEr, ASR, and SPICE where available. Results for LLaVA are in Table 11, Qwen3-VL in Table 12, and Otter together with InternVL in Table 13. Per-attack VQA results on ScienceQA are reported in Table 14.

Table 11. Backdoor-robust instruction-tuning on LLaVA. For each attack and dataset, we report CIDEr (higher is better), attack success rate (ASR; lower is better), and SPICE (higher is better).

Defense	Attack	Image Captioning						Spot the Difference		
		COCO			Flickr			CGD		
		CIDEr $\uparrow$	ASR(%) $\downarrow$	SPICE $\uparrow$	CIDEr $\uparrow$	ASR(%) $\downarrow$	SPICE $\uparrow$	CIDEr $\uparrow$	ASR(%) $\downarrow$	SPICE $\uparrow$
No defense	BadNets	75.6	1.7	25.1	52.2	0.5	20.6	146.2	100	45.4
	Blend	71.1	76.3	25.2	48.3	91.1	20.8	145.1	100	45.1
	DualKey	77.1	44.4	25.1	52.5	59.5	20.8	144.3	100	45.7
	VL-Trojan	75.9	69.7	25.1	52.7	78.1	20.8	143.9	100	45.3
ONION (Qi et al., 2021)	BadNets	75.6	1.7	–	52.2	0.1	–	146.2	100	45.4
	Blend	71.1	76.3	–	48.3	91.1	–	145.1	100	45.1
	DualKey	76.5	3.1	–	52.5	0.5	–	146.4	0.1	45.7
	VL-Trojan	75.9	1.6	–	52.2	0.3	–	147.3	1.5	45.3
BYE (Rong et al., 2025)	BadNets	80.9	1.5	–	54.4	0.2	–	145.3	100	44.8
	Blend	77.2	46.7	–	51.7	69.4	–	146.5	100	45.1
	DualKey	81.2	79.6	–	55.5	83.5	–	138.5	100	44.4
	VL-Trojan	83.1	91.4	–	57.1	96.3	–	146.1	100	45.6
BYORn-F	BadNets	61.5	1.4	25.4	42.6	0.4	20.9	130.4	1.3	43.8
	Blend	60.3	1.6	25.3	42.5	0.4	20.9	130.2	1.4	43.8
	DualKey	60.1	1.5	25.3	42.2	0.3	20.9	129.3	0	43.9
	VL-Trojan	60.6	1.8	25.2	42.1	0.2	21.0	129.2	1.1	43.8
BYORn	BadNets	90.1	1.5	22.1	61.0	0.1	18.0	134.1	1.4	44.2
	Blend	90.8	0.0	22.2	62.3	0	18.0	133.7	1.4	44.3
	DualKey	89.9	0.0	22.2	62.1	0.0	18.3	136.2	0.1	44.3
	VL-Trojan	92.9	0.0	22.7	62.5	0.0	18.2	135.6	1.4	44.2

Table 12. Backdoor-robust instruction-tuning on Qwen3-VL. For each attack and dataset, we report CIDEr (higher is better), attack success rate (ASR; lower is better), and SPICE (higher is better).

Defense	Attack	Image Captioning						Spot the Difference		
		COCO			Flickr			CGD		
		CIDEr $\uparrow$	ASR(%) $\downarrow$	SPICE $\uparrow$	CIDEr $\uparrow$	ASR(%) $\downarrow$	SPICE $\uparrow$	CIDEr $\uparrow$	ASR(%) $\downarrow$	SPICE $\uparrow$
No defense	BadNets	67.9	88.1	23.8	62.1	100	24.7	160.8	99.9	46.0
	Blend	64.5	87.8	23.5	60.4	100	24.3	159.1	98.6	46.3
	DualKey	67.9	86.7	23.6	59.8	100	24.4	154.4	99.6	45.8
	VL-Trojan	67.3	96.5	23.4	60.5	100	24.2	158.9	99.7	46.1
ONION (Qi et al., 2021)	BadNets	67.9	88.1	–	62.1	100	–	160.8	99.9	46.0
	Blend	64.5	87.8	–	60.4	100	–	159.1	98.6	46.3
	DualKey	67.9	6.1	–	59.5	3.9	–	155.9	0.8	45.9
	VL-Trojan	67.3	1.7	–	60.3	0	–	160.2	1.2	46.1
BYE (Rong et al., 2025)	BadNets	63.1	75.2	–	57.3	100	–	158.2	99.6	46.6
	Blend	63.4	97.8	–	57.7	99	–	156.0	96.6	46.0
	DualKey	63.6	97.1	–	57.9	100	–	158.1	100	46.6
	VL-Trojan	63.5	97.2	–	57.7	100	–	151.0	99.9	45.9
BYORn-F	BadNets	63.1	1.5	23.4	52.1	0	24.1	149.5	1.1	45.4
	Blend	62.8	1.5	23.4	52.2	0	24.1	150.2	1.0	45.4
	DualKey	62.4	4.5	23.5	52.3	3.4	24.0	149.2	0.0	45.3
	VL-Trojan	63.3	1.6	23.4	52.1	0	24.3	148.8	1.0	45.4
BYORn	BadNets	57.9	1.5	23.6	51.5	0.1	24.1	149.6	1.0	45.9
	Blend	59.9	1.6	23.5	51.9	0.1	24.0	148.4	1.0	45.4
	DualKey	61.3	5.9	23.3	53.9	8.1	24.1	149.0	0.1	45.4
	VL-Trojan	59.4	1.6	23.7	51.4	0	24.2	146.9	1.1	45.6

Table 13. Backdoor-robust instruction-tuning on Otter and InternVL. For each attack and dataset, we report CIDEr (higher is better) and attack success rate (ASR; lower is better).

Defense	Attack	Otter				InternVL			
		COCO		Flickr		COCO		Flickr	
		CIDEr	ASR	CIDEr	ASR	CIDEr	ASR	CIDEr	ASR
No defense	BadNets	63.0	98.4	41.4	99.9	64.5	59.5	57.6	99.9
	Blend	63.5	99.6	40.2	99.5	64.5	72.2	57.2	99.7
	DualKey	63.9	99.8	40.7	100	64.7	87.0	57.3	99.8
	VL-Trojan	65.5	100	39.9	100	65.8	67.8	57.0	99.8
ONION (Qi et al., 2021)	BadNets	63.0	98.4	41.4	99.9	64.5	59.5	57.6	99.9
	Blend	63.5	99.6	40.2	99.5	64.5	72.2	57.2	99.7
	DualKey	64.8	2.3	42.4	0.1	64.7	9.0	58.2	4.0
	VL-Trojan	63.7	8.3	40.7	11.6	65.8	1.6	58.6	0.1
BYE (Rong et al., 2025)	BadNets	61.9	100	40.9	100	68.6	72.3	61.5	99.1
	Blend	60.8	100	38.4	100	66.3	95.3	60.3	99.8
	DualKey	64.0	100	40.5	100	65.7	7.3	66.4	6.3
	VL-Trojan	63.1	100	42.8	100	65.4	81.2	61.1	100
BYORn-F	BadNets	59.5	1.2	39.4	0.2	66.9	1.7	60.7	0
	Blend	58.8	0.2	41.3	0	66.8	1.4	60.6	0.1
	DualKey	57.8	2.2	40.3	0	67.1	5.8	61.3	2.4
	VL-Trojan	56.8	1.1	38.7	0	66.3	1.4	60.4	0
BYORn	BadNets	58.6	1.3	38.8	0	68.3	1.5	57.4	0
	Blend	56.8	0	39.5	0	64.5	1.5	55.7	0.1
	DualKey	56.7	1.0	37.6	0	63.8	6.8	54.4	3.1
	VL-Trojan	59.1	0	38.1	0	63.9	1.5	56.4	0

Table 14. Backdoor-robust instruction-tuning on ScienceQA (VQA). For each attack, we report classification accuracy (ACC; higher is better) and attack success rate (ASR; lower is better).

Defense	Attack	Qwen3-VL		LLaVA-1.5	
		ScienceQA		ScienceQA	
		ACC $\uparrow$	ASR $\downarrow$	ACC $\uparrow$	ASR $\downarrow$
No defense	BadNets	89.7	61.3	80.2	100
	Blend	89.6	72.7	80.6	100
	DualKey	89.3	92.0	79.9	100
	VL-Trojan	89.9	92.4	80.4	100
ONION (Qi et al., 2021)	BadNets	89.7	61.3	80.2	100
	Blend	89.6	72.7	80.6	100
	DualKey	89.3	0.1	79.9	0
	VL-Trojan	89.9	0.1	80.4	0
BYE (Rong et al., 2025)	BadNets	90.5	0	77.2	98.6
	Blend	90.9	75.9	76.8	100
	DualKey	89.5	74.5	76.5	100
	VL-Trojan	90.3	0	74.7	100
BYORn-F	BadNets	88.9	0	68.3	0
	Blend	89.2	0	68.4	0
	DualKey	88.4	0	67.8	0
	VL-Trojan	89.1	0	67.4	0
BYORn	BadNets	86.7	0	64.6	0
	Blend	86.7	0	65.5	0
	DualKey	87.2	0	66.5	0
	VL-Trojan	86.8	0	65.4	0

D. Adaptive Attacks

We introduce an adaptive backdoor attack that leverages model-specific perturbations computed via gradient descent. This attack targets our perplexity-based detector: by finding a perturbation that makes the pretrained model assign high likelihood to the poisoned response, the poisoned sample is pushed below the detection threshold and evades filtering.

Given a pretrained victim model f_θ and a clean dataset $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{q}_i, \mathbf{y}_i)\}_{i=1}^N$, we compute sample-specific perturbations δ_i using Projected Gradient Descent (PGD), where $\mathbf{y}_i$ denotes the poisoned target response:

$$\delta_i = \arg \min_{\|\delta\|_\infty \leq \epsilon} \mathcal{L}(\theta | \mathbf{x}_i + \delta, \mathbf{q}_i, \mathbf{y}_i) \quad (10)$$

The optimization proceeds iteratively:

$$\delta^{(t+1)} = \Pi_\epsilon \left(\delta^{(t)} - \alpha \cdot \text{sign} \left(\nabla_{\delta} \mathcal{L}(\theta | \mathbf{x}_i + \delta^{(t)}, \mathbf{q}_i, \mathbf{y}_i) \right) \right) \quad (11)$$

where α is the step size, Π_ϵ denotes projection onto the ϵ -ball, and $\mathcal{L}$ is the detection score defined in Eq. 3.

We use $\epsilon = 0.09$, $\alpha = 0.01$, and $T = 10$ PGD iterations. Perturbations are computed in CLIP-normalized image space to ensure consistency during training and inference.

Training. We select 10% of training samples as poison candidates. For each, we load its pre-computed perturbation, apply it to the image, and replace the ground truth response with the target response.

Evaluation. Perturbations are generated for all test samples using the trained model. We measure Attack Success Rate (ASR) as the fraction of perturbed samples where the model outputs the target response.

E. Hyperparameter sensitivity

We validate hyperparameters using LLaVA, instruction-tuned on the LADD dataset, and evaluate on Flickr30k at a poisoning rate of 10%. Tables 15–17 extend the percentile threshold analysis across three attacks. The effect of EMA decay λ is reported in Table 18.

Table 15. Effect of percentile threshold p under the Blend attack (LLaVA, LADD/Flickr30k). P and R denote precision and recall. #FN are poisoned examples classified as clean; #FP are clean examples classified as suspicious.

Method	p	CIDEr $\uparrow$	ASR $\downarrow$	SPICE $\uparrow$	P	R	#FN	#FP
BYORn	0.3	55.6	0	19.5	0.14	1	0	13944
BYORn	0.5	55.2	0	16.7	0.20	1	0	9296
BYORn	0.6	62.2	0	18.0	0.25	1	0	6972
BYORn	0.9	61.4	0	18.2	0.99	0.99	3	3
BYORn-F	0.95	52.4	57.6	18.9	1	0.5	1162	0
BYORn	0.95	60.4	1.1	18.9	1	0.5	1162	0
BYORn-F	0.975	53.7	79.3	20.1	1	0.25	1743	0
BYORn	0.975	59.8	1.2	20.1	1	0.25	1743	0

Table 16. Effect of percentile threshold p under the DualKey attack (LLaVA, LADD/Flickr30k). P and R denote precision and recall. #FN are poisoned examples classified as clean; #FP are clean examples classified as suspicious.

Method	p	CIDEr $\uparrow$	ASR $\downarrow$	SPICE $\uparrow$	P	R	#FN	#FP
BYORn	0.6	62.1	0	18.3	0.25	1	0	6972
BYORn	0.9	62.5	0	19.0	0.999	0.999	1	1
BYORn-F	0.95	54.9	0.5	19.8	1	0.5	1162	0
BYORn	0.95	62.5	0.1	18.9	1	0.5	1162	0
BYORn-F	0.975	56.7	30.7	19.9	1	0.25	1743	0
BYORn	0.975	62.0	0.3	19.9	1	0.25	1743	0

Table 17. Effect of percentile threshold p under the VL-Trojan attack (LLaVA, LADD/Flickr30k). P and R denote precision and recall. #FN are poisoned examples classified as clean; #FP are clean examples classified as suspicious.

Method	p	CIDEr $\uparrow$	ASR $\downarrow$	SPICE $\uparrow$	P	R	#FN	#FP
BYORn	0.6	62.9	0	18.3	0.25	1	0	6972
BYORn	0.9	61.6	0.1	19.3	0.99	0.99	3	3
BYORn-F	0.95	54.2	11.0	20.2	1	0.5	1162	0
BYORn	0.95	62.2	0.4	18.6	1	0.5	1162	0
BYORn-F	0.975	54.6	64.8	19.9	1	0.25	1743	0
BYORn	0.975	60.6	2.4	19.3	1	0.25	1743	0

Table 18. Effect of EMA decay λ .

λ	CIDEr	ASR
0.99	61.8	0
0.995	62.2	0
0.999	61.2	0